COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structures & Algorithms	**SECTION:**	CPE21S2
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ACTIVITY NO	**Seatwork 8.1 Using Sorting Algorithms 2**

1. PROCEDURE OUTPUT

Answer the following questions:

1. Explain the quick sort, shell sort, and merge sort types of sorting algorithms.

Quick - This sort uses a pivot that moves the elements depending on the condition where the elements less than pivot are at its left side, and the elements greater than the pivot are arranged at its right side.

Shell - Shell sorting works identically to that of "insertion sort", but instead of comparing only adjacent elements, it compares elements that are far apart from each other using a "gap" sequence. Where in most cases it uses ArraySize / 2 to determine the gap, then using the insertion sort on the elements separated by the gap, it sorts it one by one until the gap becomes 1. It then repeats this process until we get a sorted array in ascending order. The best case time complexity is O(n log n) while the worst case is 0(n^2), but it depends on the gap sequence chosen.

Merge - Merge sort splits the list into halves, sorts those pieces, then stitches them back together in order.

2. Give simple sample programs in C++ that use the above sorting algorithms. Use a user input of 10 integer values in your example elements in an array to be sorted. Explain how the programs work.

Quick - Online Shopping

Quick sort can be used to arrange product prices from lowest to highest. For example, if there are 10 product prices in an array, Quick Sort chooses one value as a pivot, then places smaller values on one side and larger values on the other side. It repeats this process on the smaller groups until all are arranged from smallest to largest.

Shell - Sorting Student Scores

Shell Sort can be used to arrange 10 student scores from lowest to highest. It starts by comparing values that

are separated by a certain gap. The gap is then gradually reduced, allowing the values to become more organized. When the gap reaches 1, it works similarly to insertion sort for backtracking and finally completing the sorted array.

Merge - Arranging Monthly Expenses

Merge sort can be used to sort 10 monthly expenses from the smallest amount to the largest. It divides the amount of expenses into halves creating smaller groups, sorts each group, and then merge them all back together until the list is arranged properly.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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